MILITARY SPECIFICATION

PROCESSING UNIT, DENTAL RESINS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a processing unit suitable for processing self-curing dental resins at elevated pressure to eliminate porosity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Phila., Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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MILITARY

MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, Or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

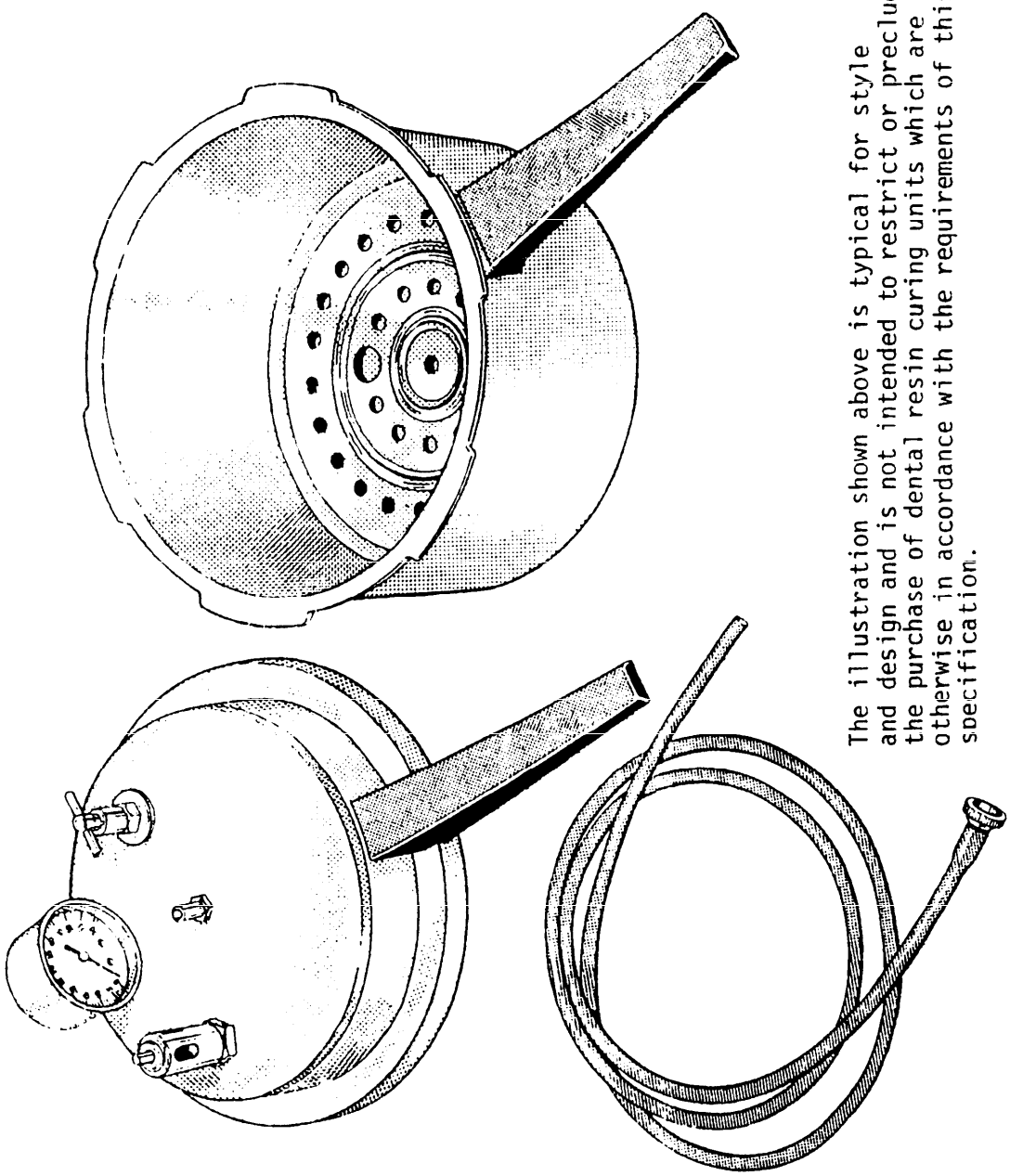
3. REQUIREMENTS

3.1 Material. Material shall be new and shall conform to the applicable requirements specified herein. Material not specifically specified shall be of high commercial quality throughout, and shall be entirely suitable for the purpose intended.

3.2 Design and construction. The processing unit shall be a heavy duty unit, as shown in figure 1, suitable for the processing of self-curing dental resins at 20 ± 5 pounds per square inch (psi), to eliminate porosity. The processing unit shall consist of a four quart, seamless, cylindrical shell having radiused corners; a two part handle; a safety type cover with a cam operating locking device; a pressure gage; an inlet valve; a pressure release valve; a rubber gasket; a spring-loaded safety valve with brass seat with Buna-N disc; and an air hose assembly for compressed air input. The furnishing of a perforated plate for holding the denture to be processed is optional.

3.2.1 Shell. The shell shall be made of one piece, seamless, deep drawn aluminum, having a minimum thickness of 0.128 inch (B&S gage no. 8). The shell shall have a handle fabricated of high temperature phenolic thermosetting plastic having a specific gravity of 1.75 to 1.90. The handle shall be of sturdy design capable of supporting the shell, completely filled with water, without loosening or breaking. The handle shall be securely fastened to the shell in such a manner which will prevent loosening or pulling free. The handle shall mate with the handle on the cover, when the cover is in the locked position. The outside dimensions of the shell shall be 9 ± 1 inches excluding the handle and 15 ± 1 inches including handle.

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The illustration shown above is typical for style and design and is not intended to restrict or preclude the purchase of dental resin curing units which are otherwise in accordance with the requirements of this specification.

FIGURE 1. Processing Unit, Dental Resin.

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3.2.2 Cover. The cover shall be rigid and fabricated of seamless aluminum having a minimum thickness of 0.126 inch. The cover shall be formed to hold a removable gasket (see 3.2) for positive leak proof seal when the unit is closed. The air inlet valve, the safety valve, the pressure gage and the pressure release valve shall each be fitted and sealed into the cover. A plastic handle shall be fabricated of the same material specified in 3.2.1 and shall be securely attached to the cover and shall mate with the handle on the shell when the unit is locked into closed position.

3.2.2.1 Safety valve. The safety valve shall be of rugged construction and suitable for the purpose intended. The valve shall be spring loaded and shall be adjusted to release pressure at 30 to 32 p.s.i., and automatically return the unit to 20 ± 5 psi. The valve shall have a brass seat with Buna-N disc.

3.2.2.2 Air inlet valve. The air inlet valve shall be a spring loaded check valve type suitable for the purpose intended.

3.2.2.3 Pressure release valve. The pressure release valve shall be a screw type quick exhaust valve, with a metal to metal seat of brass construction. The valve shall have a tee handle with a steel crossbar.

3.2.2.4 Pressure gage. The pressure gage shall be graduated for 0 to 60 pounds per square inch (psi). The minimum unit division shall be 2 psi. The pressure gage shall be of high commercial quality throughout and shall be entirely suitable for the purpose intended.

3.2.2.5 Gasket. The gasket shall be fabricated of nitrile rubber or chloroprene rubber having a hardness (durometer) of 55 to 63.

3.3.3 Air hose assembly. The unit shall be supplied with a 5 feet (minimum) length, 3/16 ± 1/64 inside diameter, 3/64 ± 1/64 inch wall thickness rubber tubing with a female threaded cap on one end for attachment to the air intake valve.

3.4 Finish. The outside surface of the aluminum shell (excluding bottom) and the outside surface of the aluminum cover shall have a polished finish. All surfaces shall be free of corrosion.

3.6 Performance. The processing unit shall be capable of maintaining an internal pressure of 20 ± 5 psi. The processing unit shall lose no more than 10 percent of its initial pressure when tested as specified in 4.4.1. The safety valve shall release the internal pressure when it rises to 30 to 32 psi and then return the internal pressure to 20 ± 5 psi when tested as specified in 4.4.2. The cover shall not be capable of being easily removed until the internal pressure has decreased below 1 psi when tested as specified in 4.4.3.

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3.7 Identification marking. The manufacturer's name or trademark shall be indicated on each unit by stamping on the bottom or by the use of a pressure sensitive label on the upper surface of the plastic handle of the cover.

3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.9 Workmanship. Workmanship shall be first class throughout. The units shall be free from defects which may affect proper functioning, appearance, or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of compliance. Certificates of compliance for the rubber gasket material, plastic handle material, and gage of aluminum may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the righ to check test such items to determine the validity of the certification.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I. The unit of product for sampling purposes shall be one complete processing unit.

TABLE I. Sampling for examination.

Type of defect	Inspection Level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5
Dimensional	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with table II. Failure of any of the tests shall be cause for rejection of the unit.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level
Internal pressure	3.6	4.4.1	100 percent
Safety valve	3.6	4.4.2	100 percent
Cover release	3.6	4.4.3	100 percent

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4.3 Examination.

4.3.1 Processing unit. The processing unit shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination however, shall not be restricted to defects listed in table III.

TABLE III. Classification of defects

Classification	Defect
Major	
101	Any components missing.
102	Any component broken or damaged.
103	Unit not free of cracks, burrs or sharp edges.
104	Pressure gage not securely attached to cover.
105	Pressure relief valve not securely attached to cover.
106	Air inlet valve not securely attached to cover.
107	Safety valve not securely attached to cover.
108	Gasket does not fit properly in access.
109	Cover does not mate properly with shell.
110	Cover dented.
111	Shell dented.
112	Gage markings not as specified.
113	Handle not securely attached to cover.
114	Handle not securely attached to shell.
115	Fittings on air hose does not mate with air inlet fitting.
Minor	
201	Finish not as specified.
202	Operating instructions missing, inaccurate, or inadequate.

4.3.2 Dimensional examination. The component parts shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the unit packaging, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Pressure loss. Each closed unit shall be filled with air until the pressure gage reads 20 ± 5 psi. The unit shall be allowed to stand under pressure for 1 hour and the gage read to determine pressure loss.

4.4.2 Safety valve. Each closed unit shall be filled with air under 30 to 32 p.s.i. and the safety valve shall be tested to determine if it actuates to release the pressure and return to that pressure specified in 3.6.

4.4.3 Cover release. Each closed unit shall be filled with air until the pressure gage reads 20 ± 5 psi. The air shall then be released until 1.0 p.s.i. remains in the unit. An attempt shall be made to open the unit.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each processing unit shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure of box shall be as specified in the appendix of PPP-B-636.

5.1.2 Level C. Processing units shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Four processing units shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Four processing units shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged processing units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES.

6.1 Intended use. The processing unit covered by this specification is intended for polymerizing self-curing resins under controlled pressure for maximum strength, density and color stability.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-890-1382	PROCESSING UNIT, DENTAL RESINS

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA - DM

Project No. 6520-1433

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-37582 PROCESSSING UNIT, DENTAL RESINS

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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